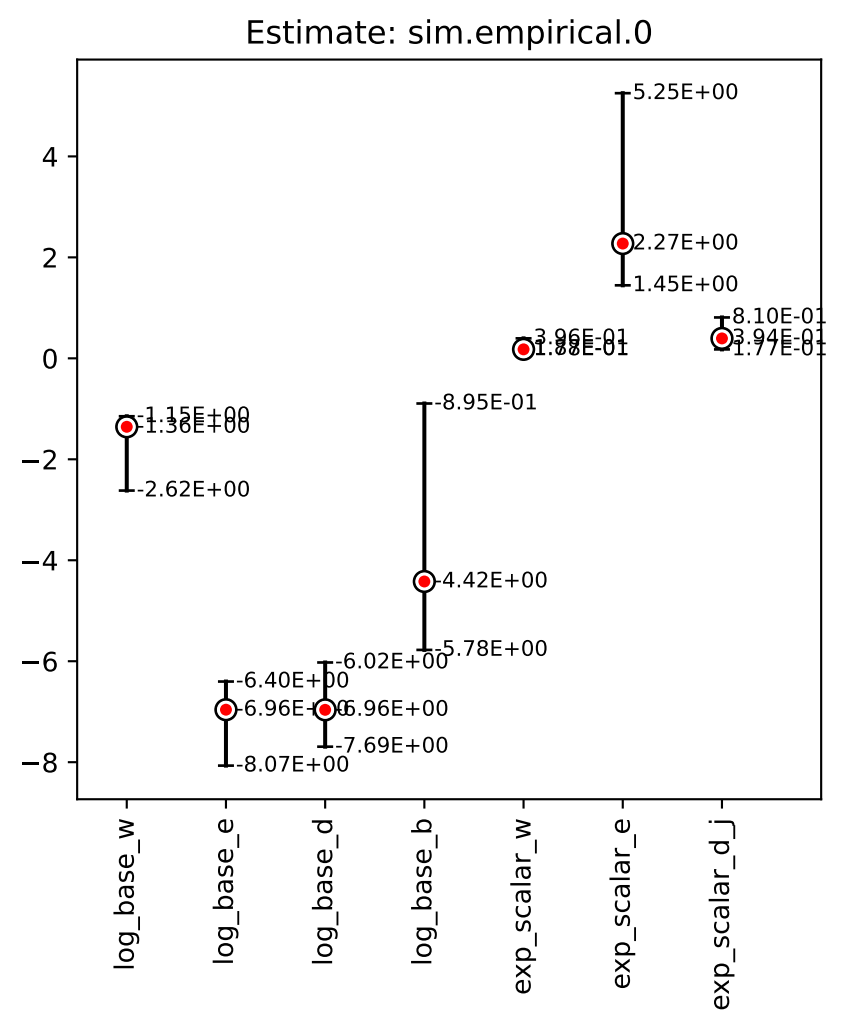
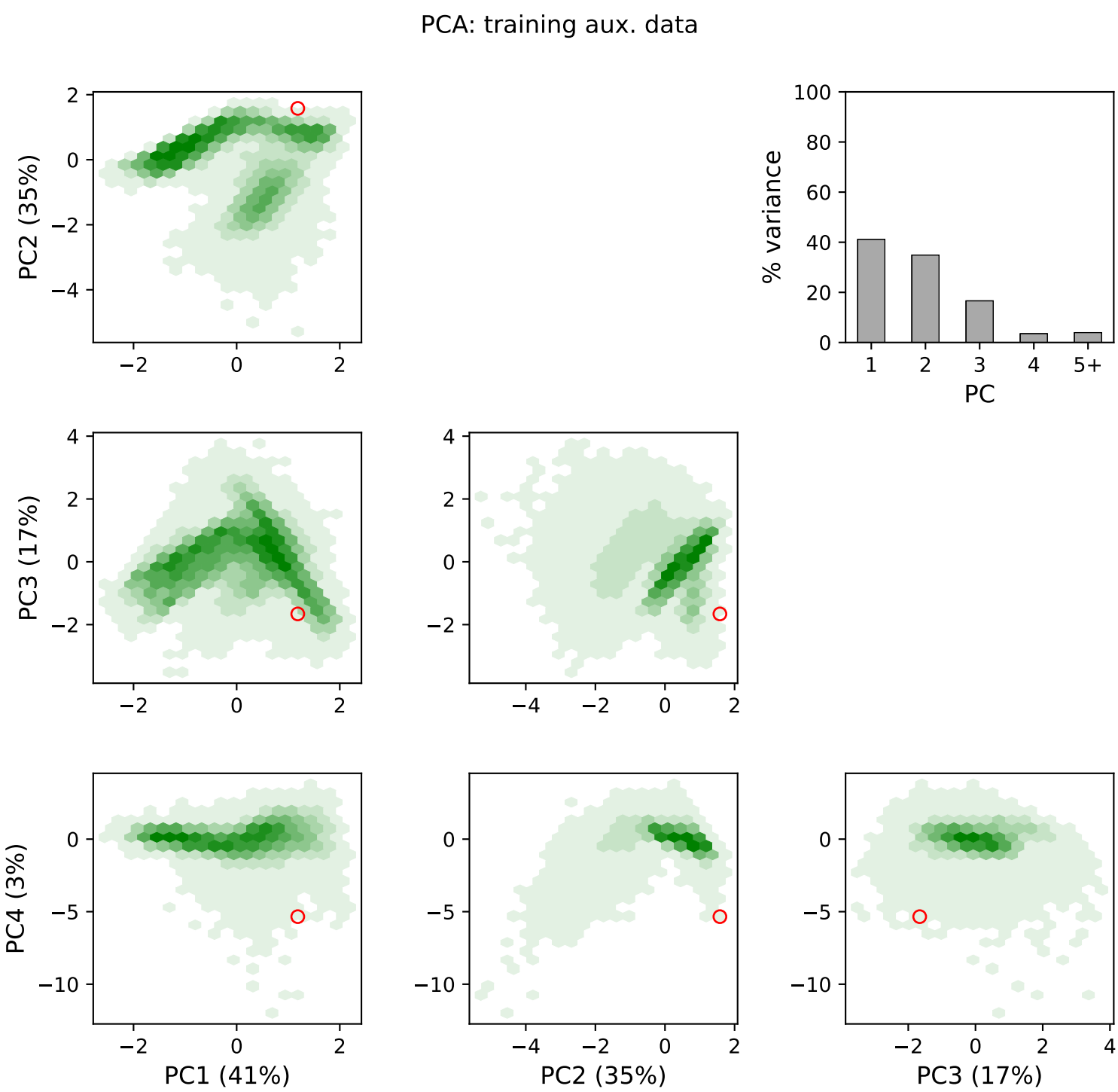
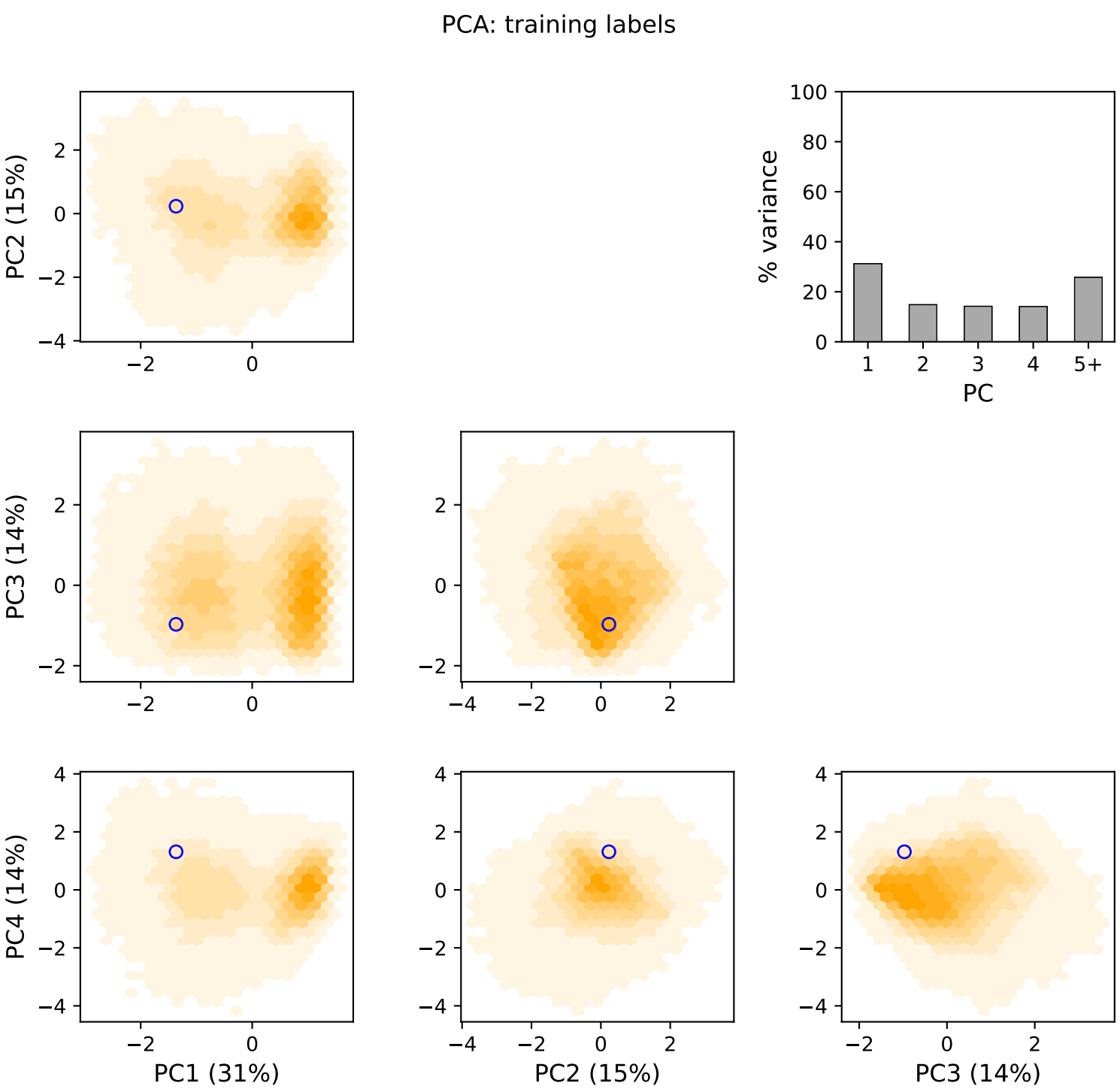
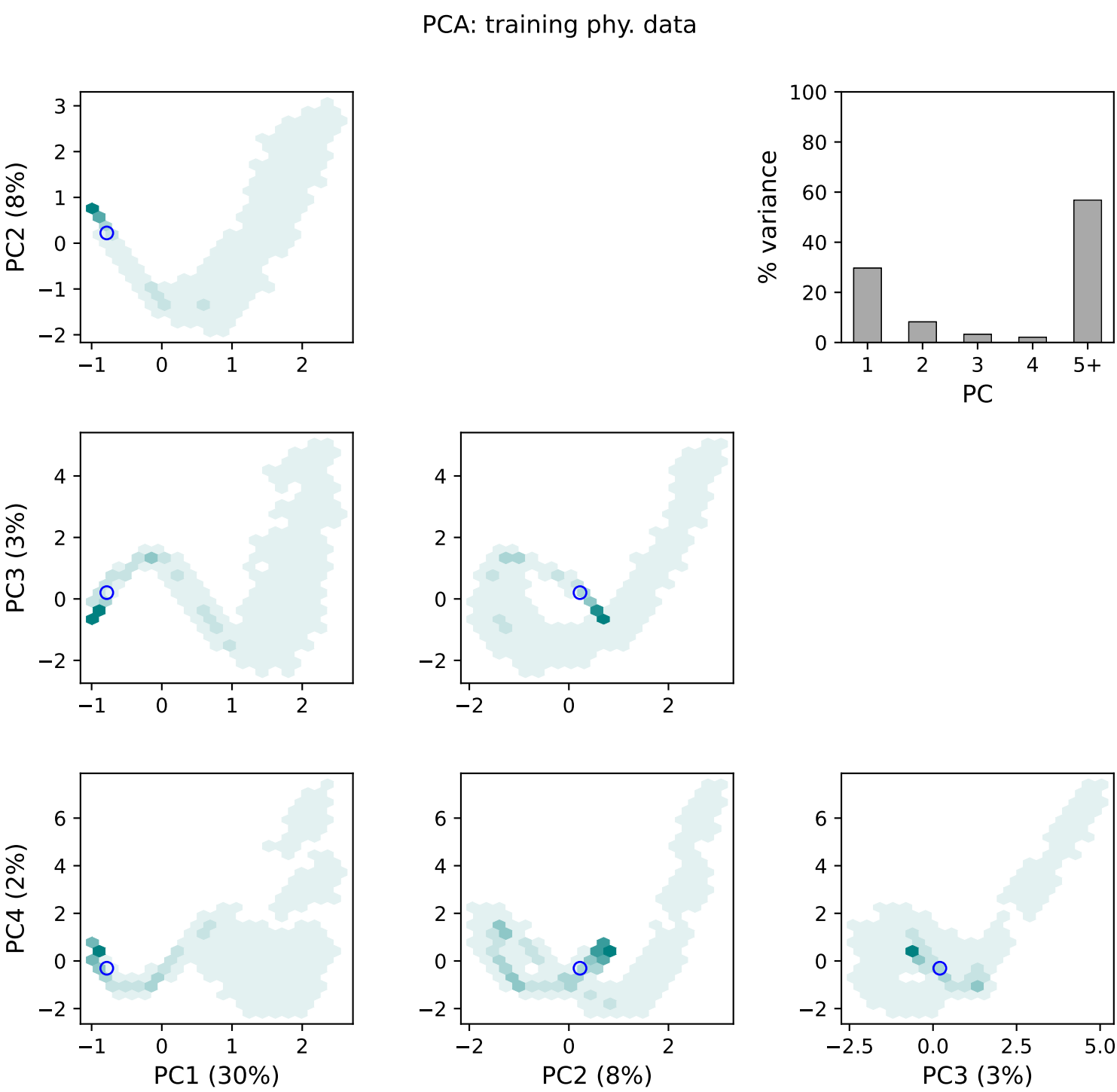
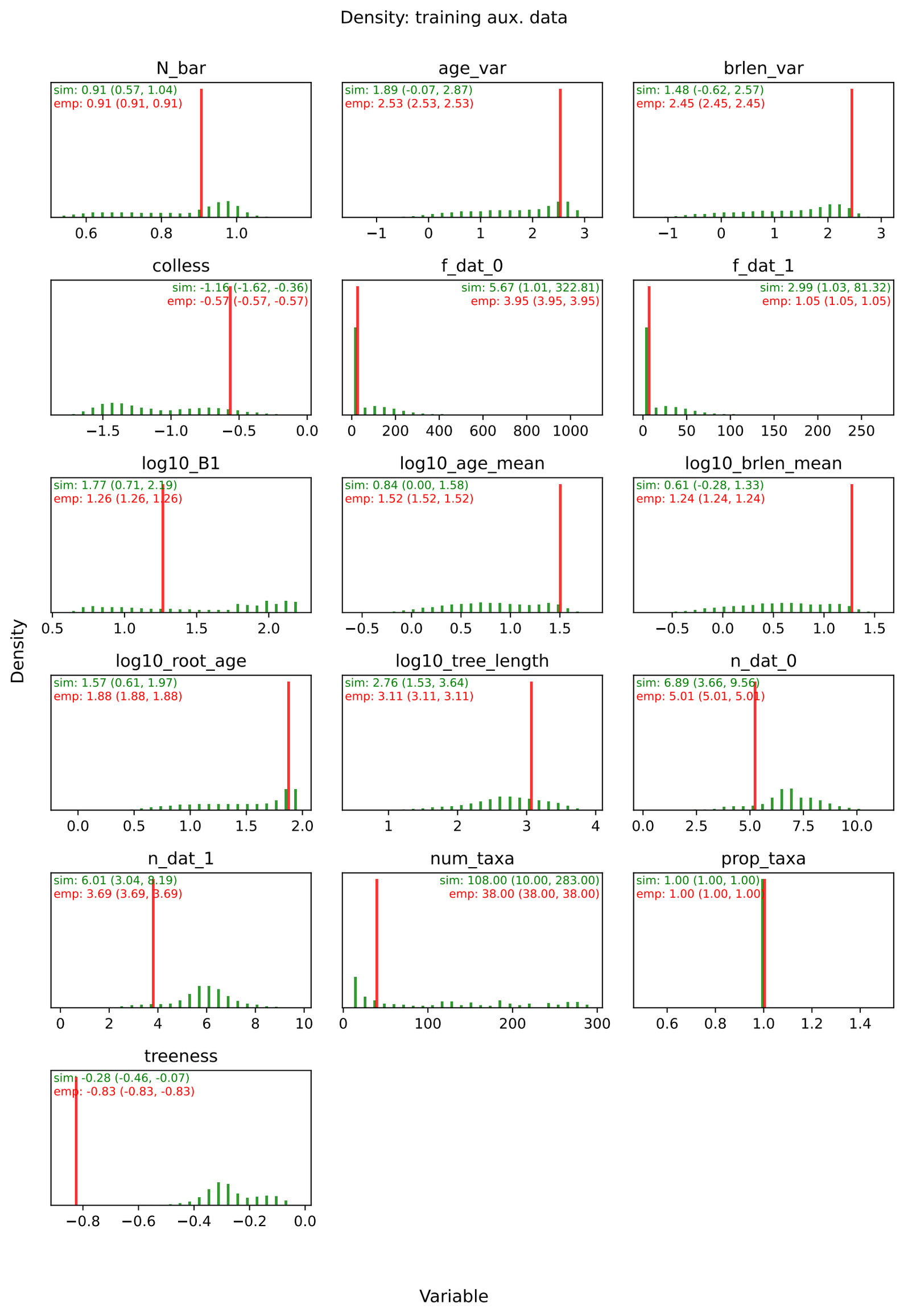
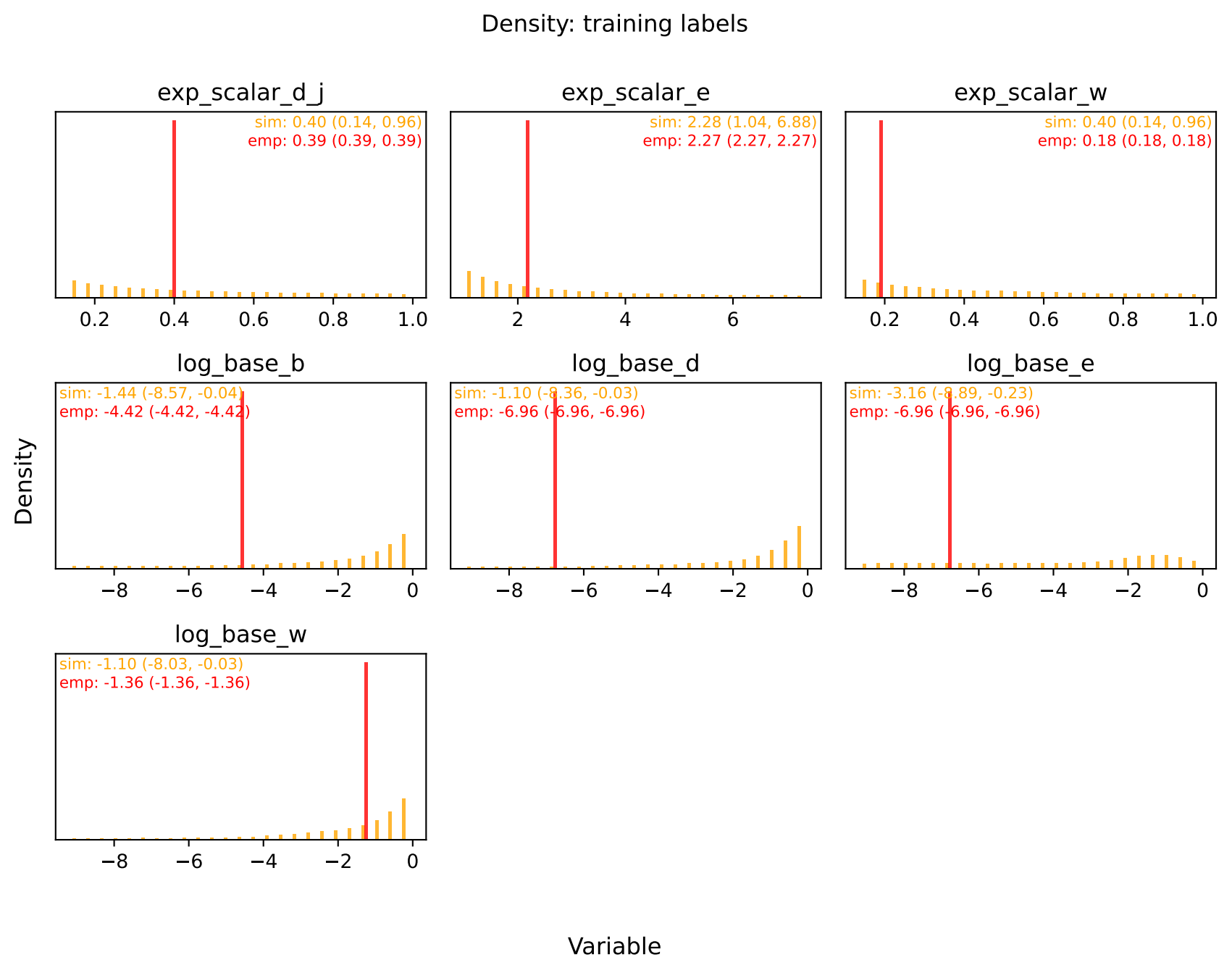

Estimate: sim.empirical.0
5.25E+00
2.27E+00
1.45E+00
8.10E-01
3.94E-01
1.77E-01
-1.15E+00
-1.36E+00
-2.62E+00
-8.95E-01
-4.42E+00
-5.78E+00
-6.40E+00
-6.96E+00
-8.07E+00
-6.02E+00
-6.96E+00
-7.69E+00
4
2
0
−2
−4
−6
−8
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w
exp_scalar_e
exp_scalar_d_j

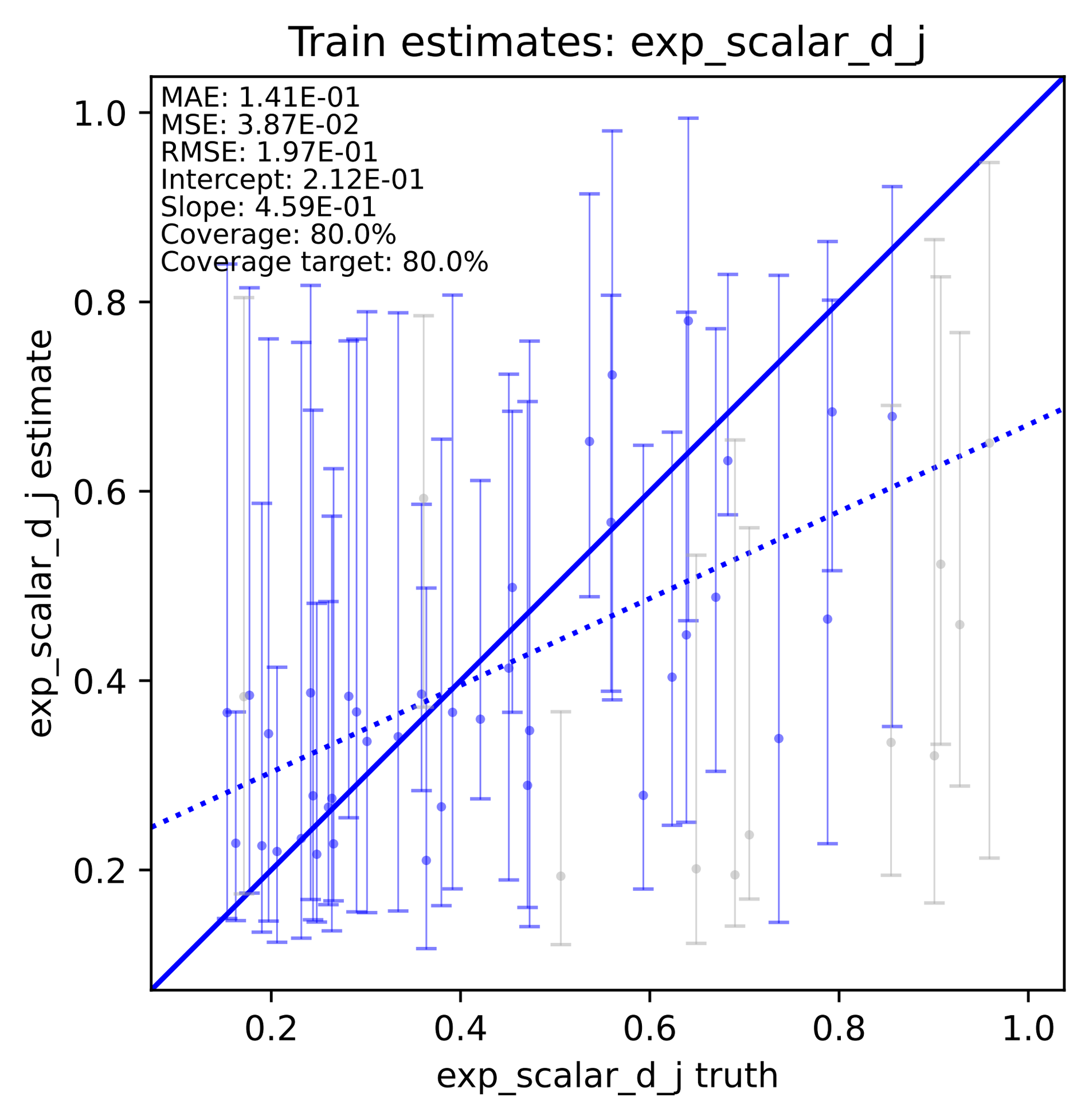

PCA: training aux. data
PC2 (35%)
% variance
PC
PC3 (17%)
PC4 (3%)
PC1 (41%)
PC2 (35%)
PC3 (17%)

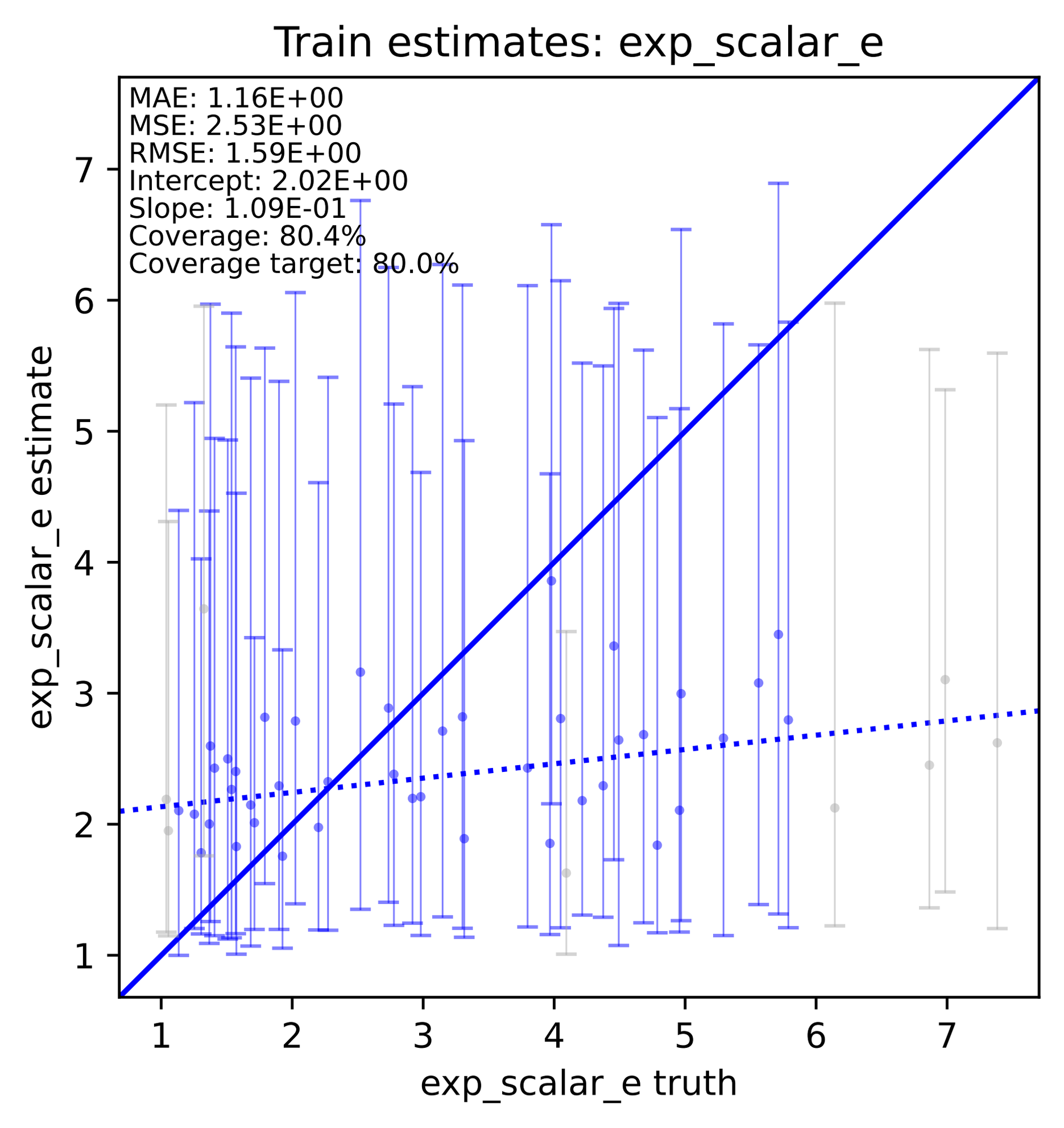

PCA: training labels
PC2 (15%)
% variance
PC
PC3 (14%)
PC4 (14%)
PC1 (31%)
PC2 (15%)
PC3 (14%)

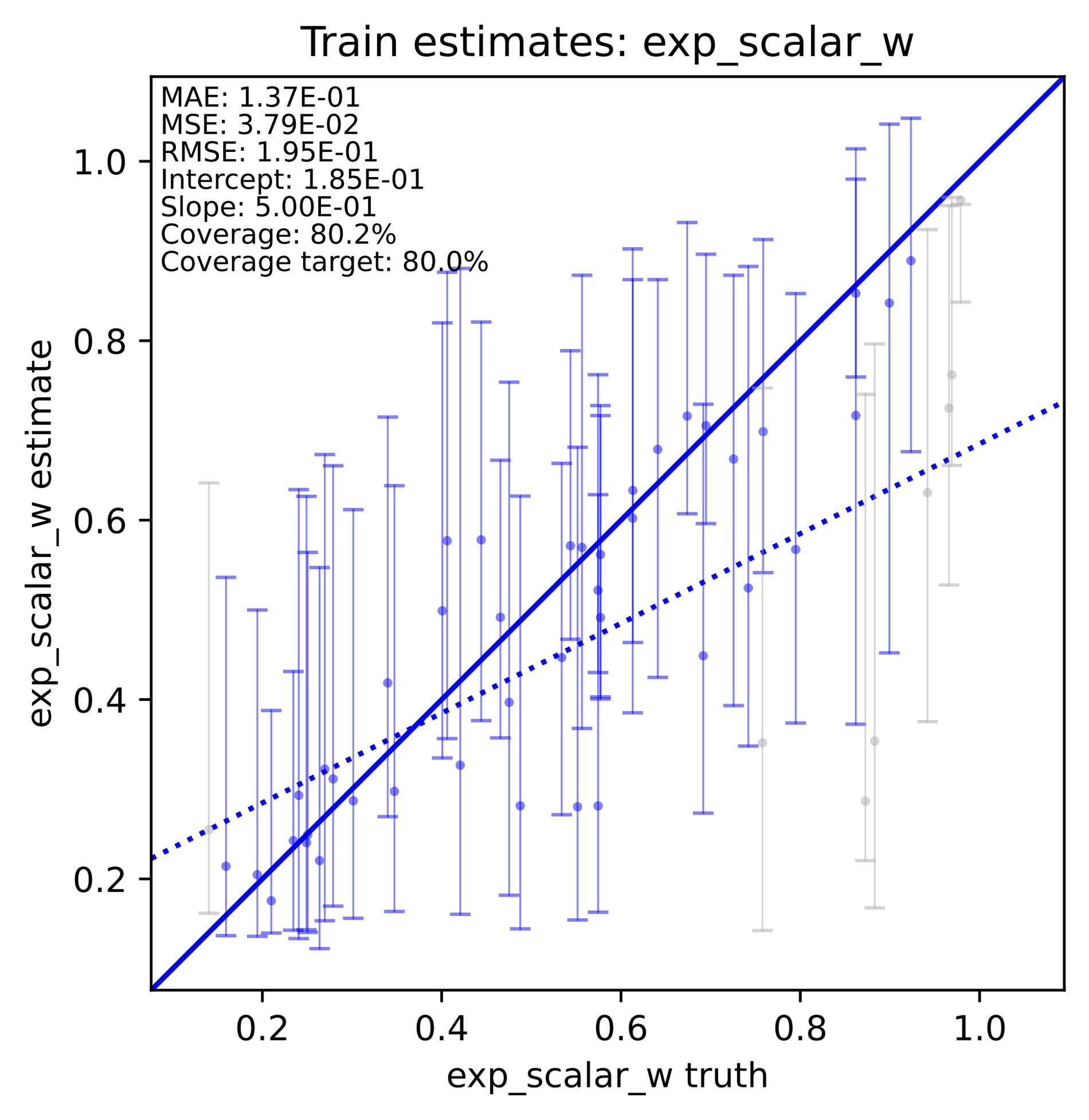

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (3%)
PC4 (2%)
PC1 (30%)
PC2 (8%)
PC3 (3%)

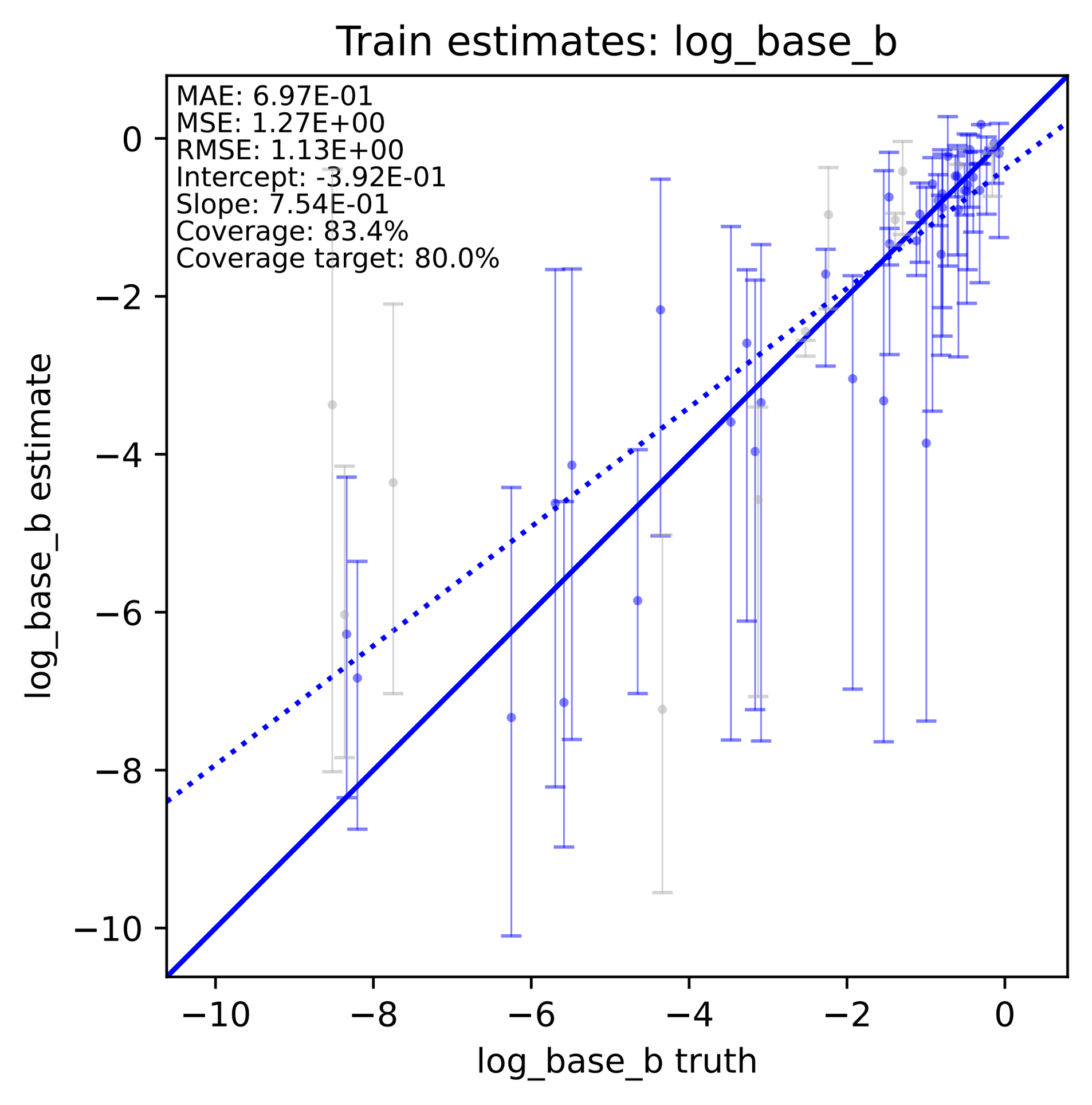

Density: training aux. data
N_bar
sim: 0.91 (0.57, 1.04)
emp: 0.91 (0.91, 0.91)
age_var
sim: 1.89 (-0.07, 2.87)
emp: 2.53 (2.53, 2.53)
brlen_var
sim: 1.48 (-0.62, 2.57)
emp: 2.45 (2.45, 2.45)
colless
sim: -1.16 (-1.62, -0.36)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 5.67 (1.01, 322.81)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.99 (1.03, 81.32)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 1.77 (0.71, 2.19)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.84 (0.00, 1.58)
emp: 1.52 (1.52, 1.52)
log10_brlen_mean
sim: 0.61 (-0.28, 1.33)
emp: 1.24 (1.24, 1.24)
log10_root_age
sim: 1.57 (0.61, 1.97)
emp: 1.88 (1.88, 1.88)
log10_tree_length
sim: 2.76 (1.53, 3.64)
emp: 3.11 (3.11, 3.11)
n_dat_0
sim: 6.89 (3.66, 9.56)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 6.01 (3.04, 8.19)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 108.00 (10.00, 283.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.46, -0.07)
emp: -0.83 (-0.83, -0.83)
Density
Variable

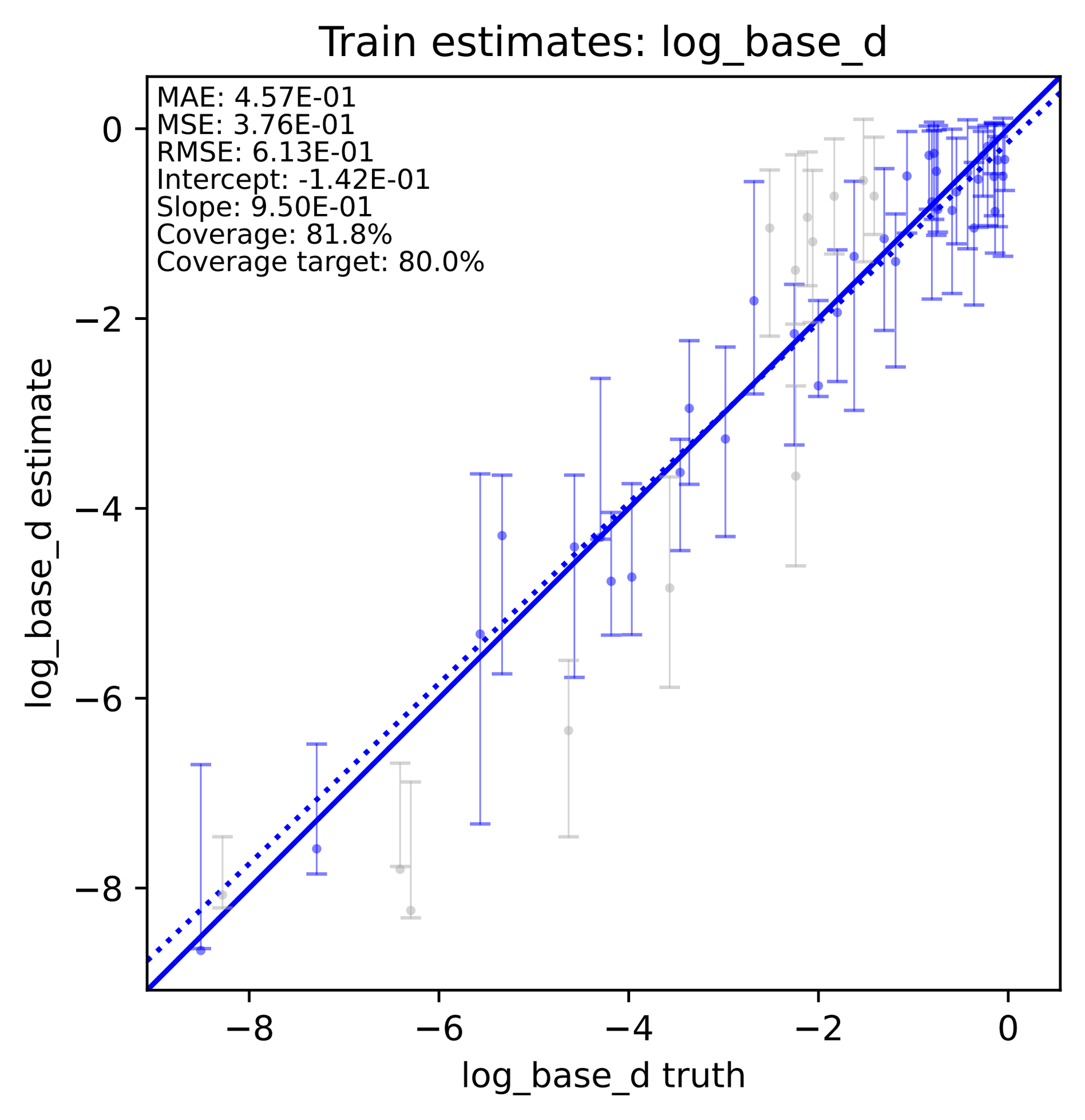

Density: training labels
exp_scalar_d_j
sim: 0.40 (0.14, 0.96)
emp: 0.39 (0.39, 0.39)
exp_scalar_e
sim: 2.28 (1.04, 6.88)
emp: 2.27 (2.27, 2.27)
exp_scalar_w
sim: 0.40 (0.14, 0.96)
emp: 0.18 (0.18, 0.18)
log_base_b
sim: -1.44 (-8.57, -0.04)
emp: -4.42 (-4.42, -4.42)
log_base_d
sim: -1.10 (-8.36, -0.03)
emp: -6.96 (-6.96, -6.96)
log_base_e
sim: -3.16 (-8.89, -0.23)
emp: -6.96 (-6.96, -6.96)
log_base_w
sim: -1.10 (-8.03, -0.03)
emp: -1.36 (-1.36, -1.36)
Density
Variable

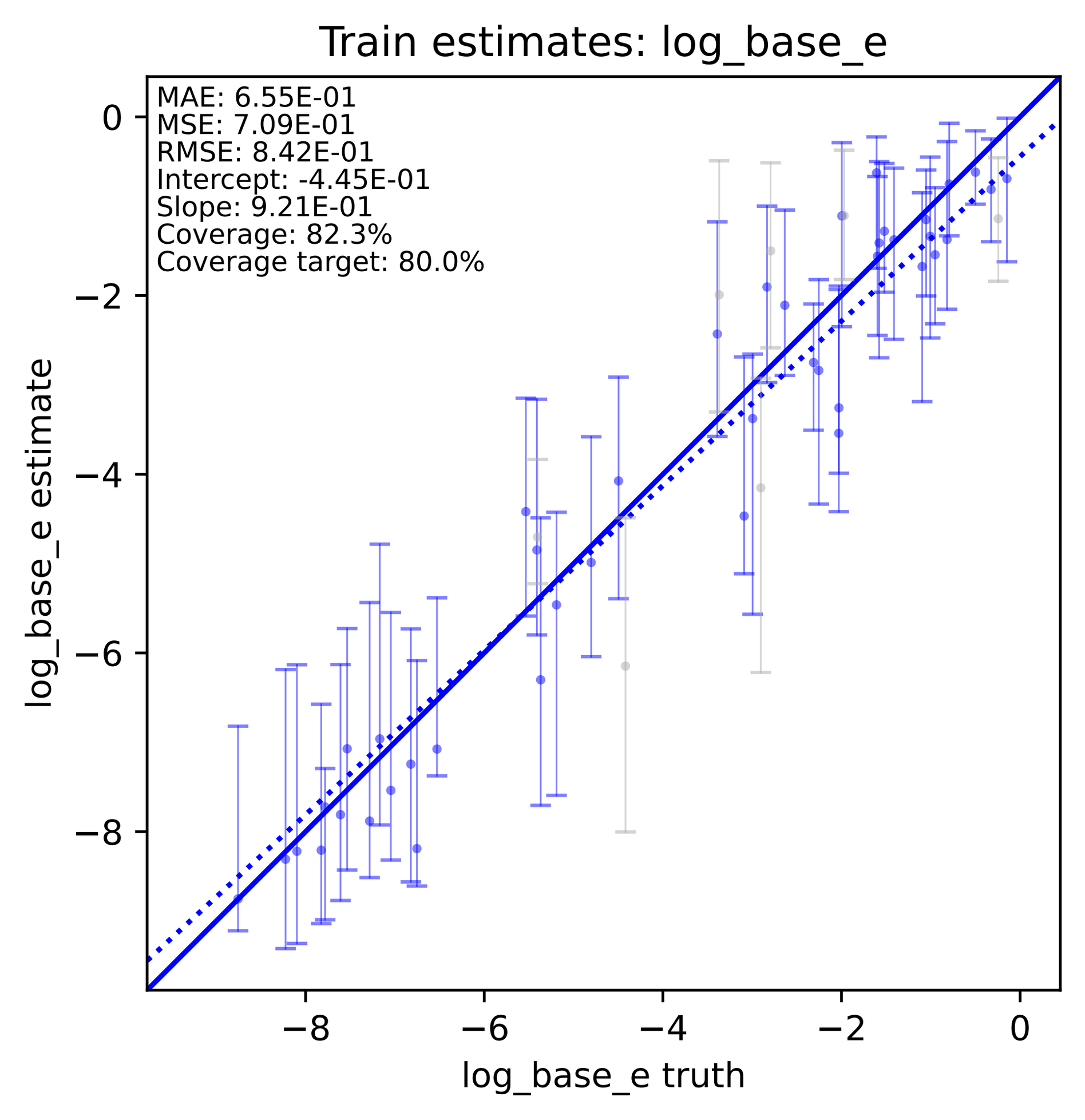

Train estimates: exp_scalar_d_j
MAE: 1.41E-01
MSE: 3.87E-02
RMSE: 1.97E-01
Intercept: 2.12E-01
Slope: 4.59E-01
Coverage: 80.0%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

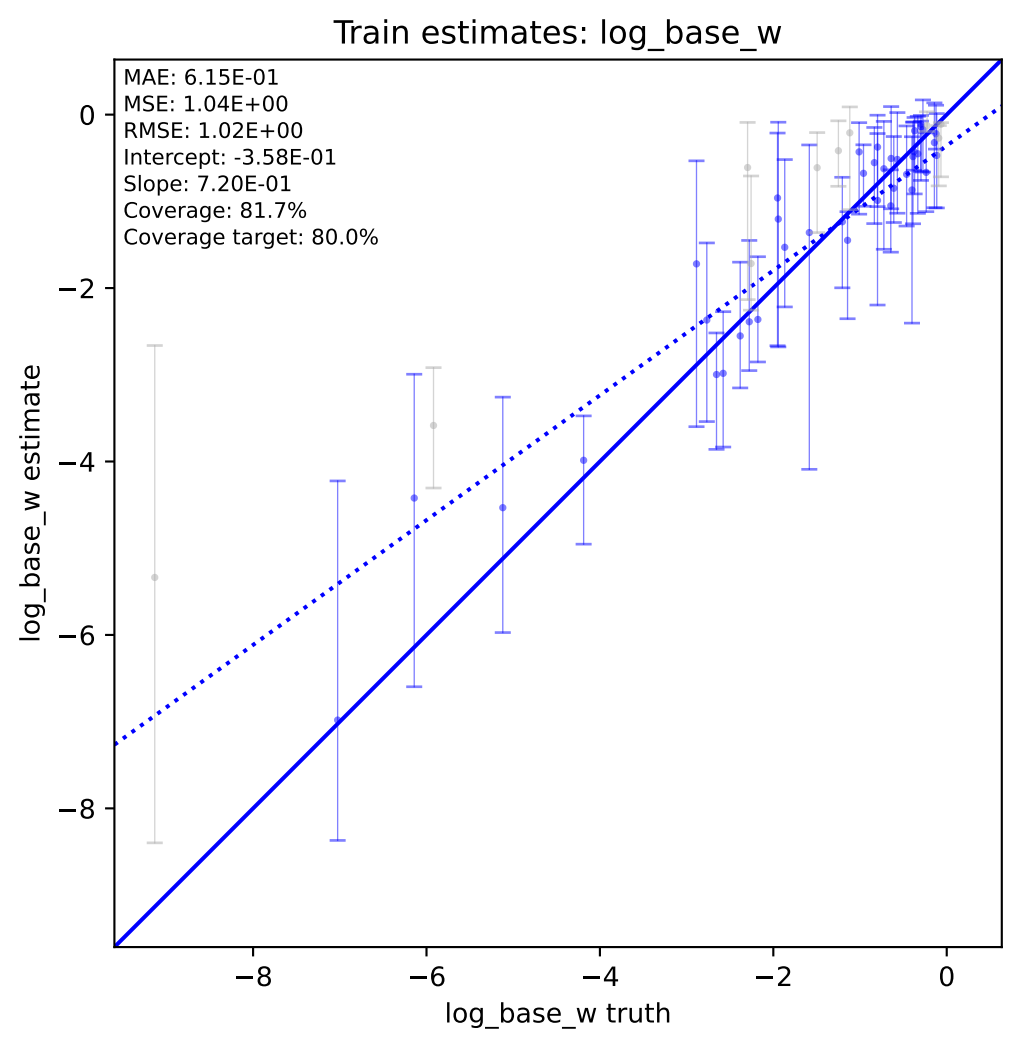

Train estimates: exp_scalar_e
MAE: 1.16E+00
MSE: 2.53E+00
RMSE: 1.59E+00
Intercept: 2.02E+00
Slope: 1.09E-01
Coverage: 80.4%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

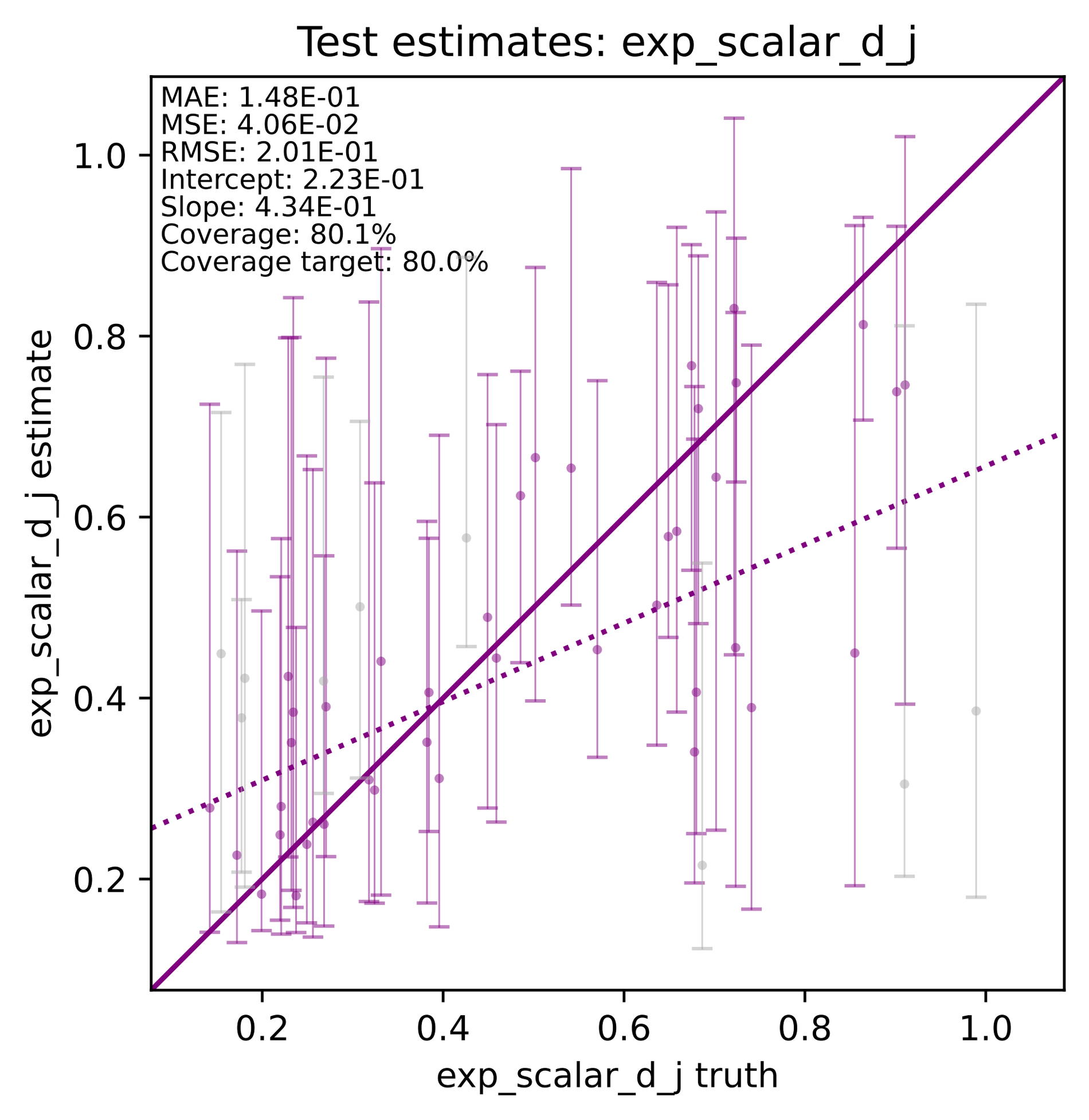

Train estimates: exp_scalar_w
MAE: 1.37E-01
MSE: 3.79E-02
RMSE: 1.95E-01
Intercept: 1.85E-01
Slope: 5.00E-01
Coverage: 80.2%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

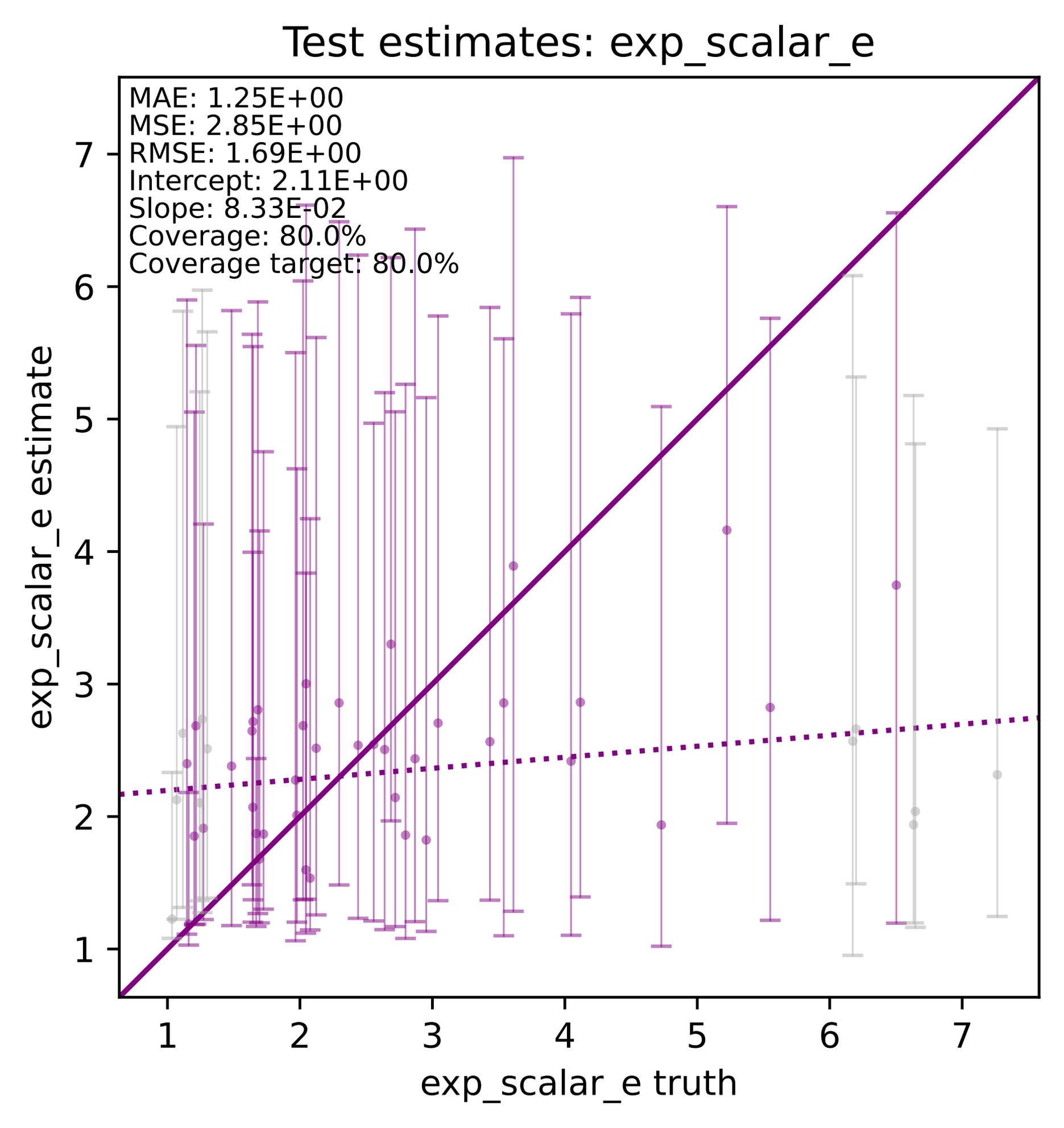

Train estimates: log_base_b
MAE: 6.97E-01
MSE: 1.27E+00
RMSE: 1.13E+00
Intercept: -3.92E-01
Slope: 7.54E-01
Coverage: 83.4%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

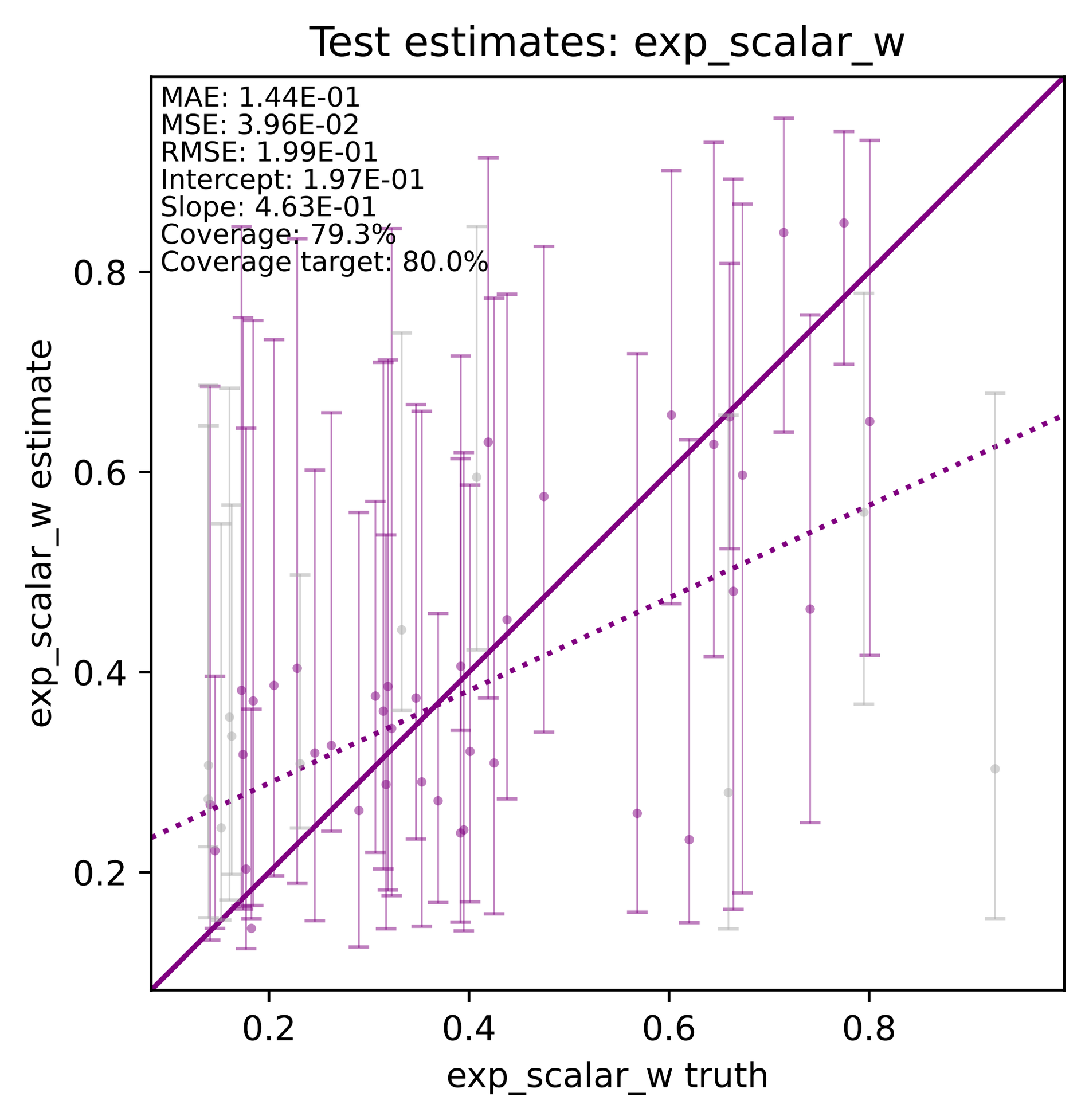

Train estimates: log_base_d
MAE: 4.57E-01
MSE: 3.76E-01
RMSE: 6.13E-01
Intercept: -1.42E-01
Slope: 9.50E-01
Coverage: 81.8%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

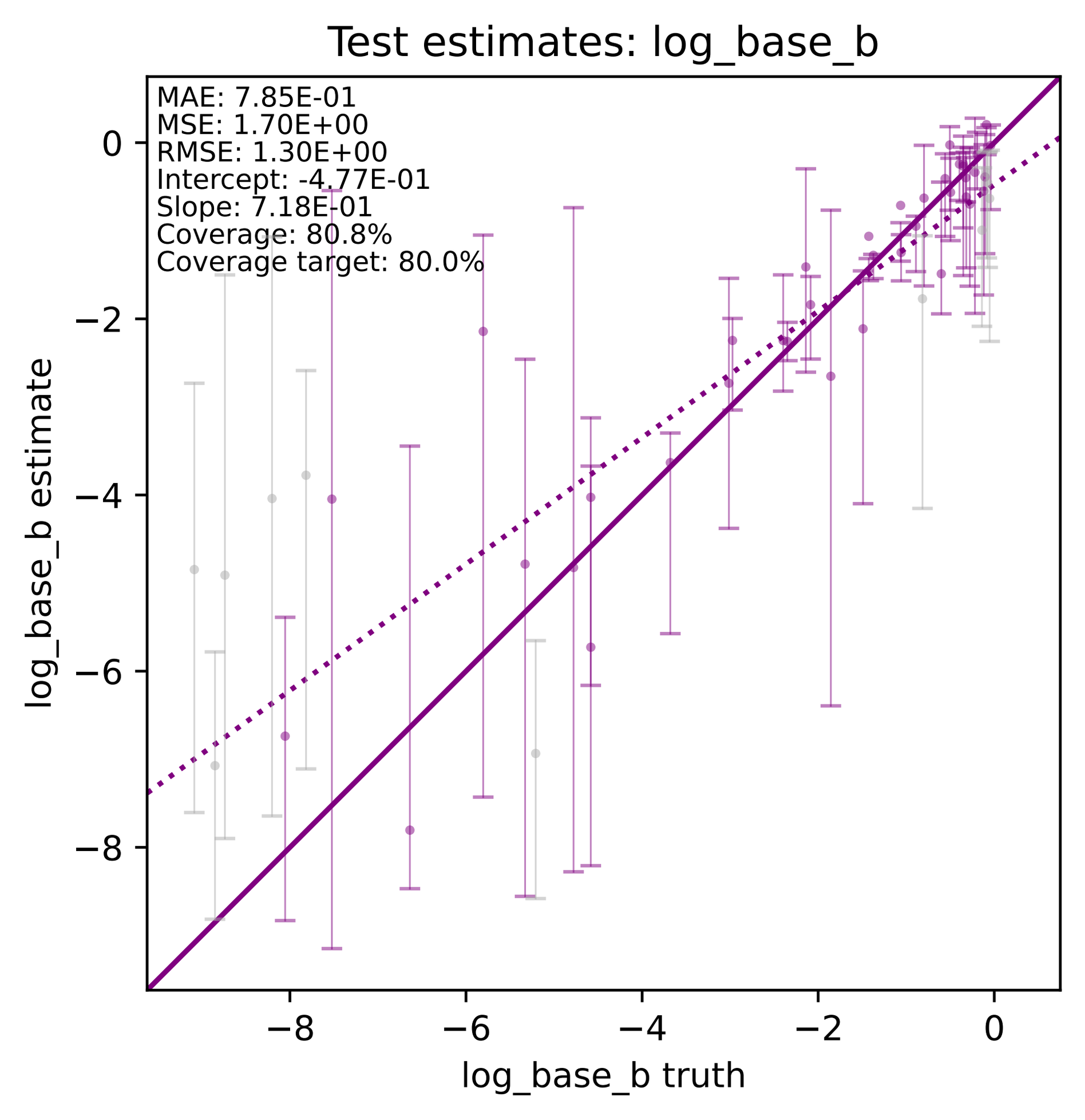

Train estimates: log_base_e
MAE: 6.55E-01
MSE: 7.09E-01
RMSE: 8.42E-01
Intercept: -4.45E-01
Slope: 9.21E-01
Coverage: 82.3%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

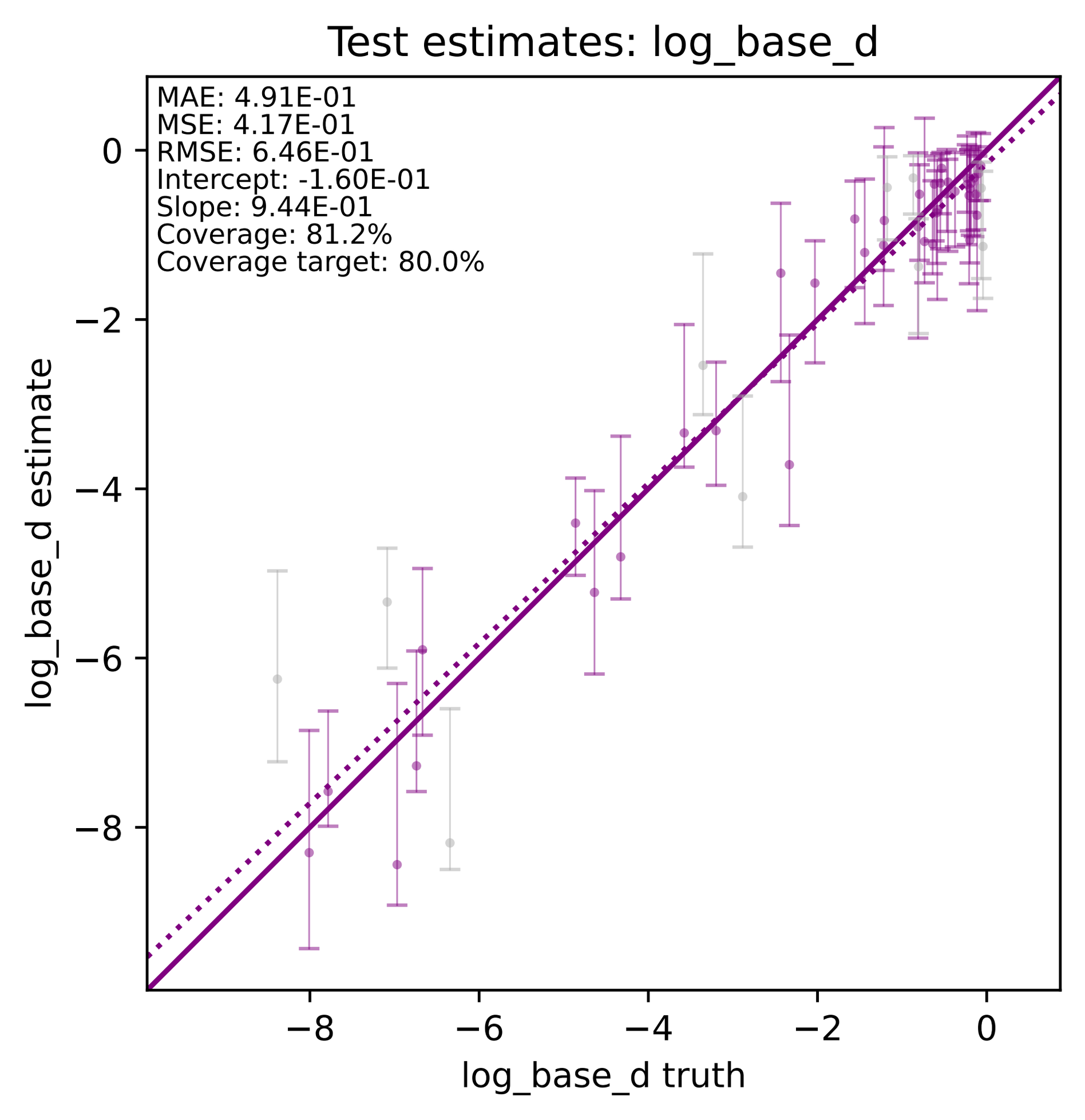

Train estimates: log_base_w
MAE: 6.15E-01
MSE: 1.04E+00
RMSE: 1.02E+00
Intercept: -3.58E-01
Slope: 7.20E-01
Coverage: 81.7%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8

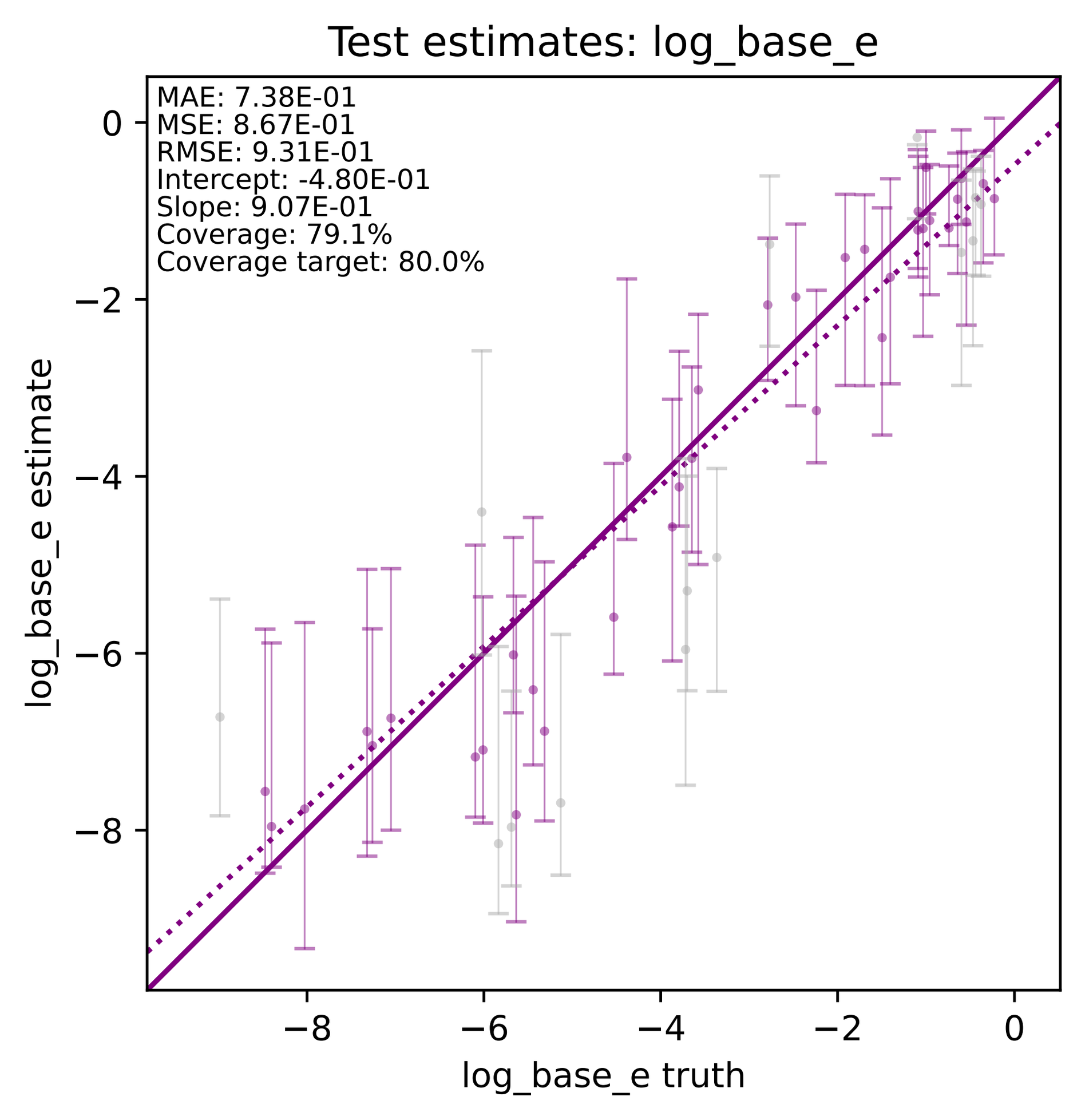

Test estimates: exp_scalar_d_j
MAE: 1.48E-01
MSE: 4.06E-02
RMSE: 2.01E-01
Intercept: 2.23E-01
Slope: 4.34E-01
Coverage: 80.1%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

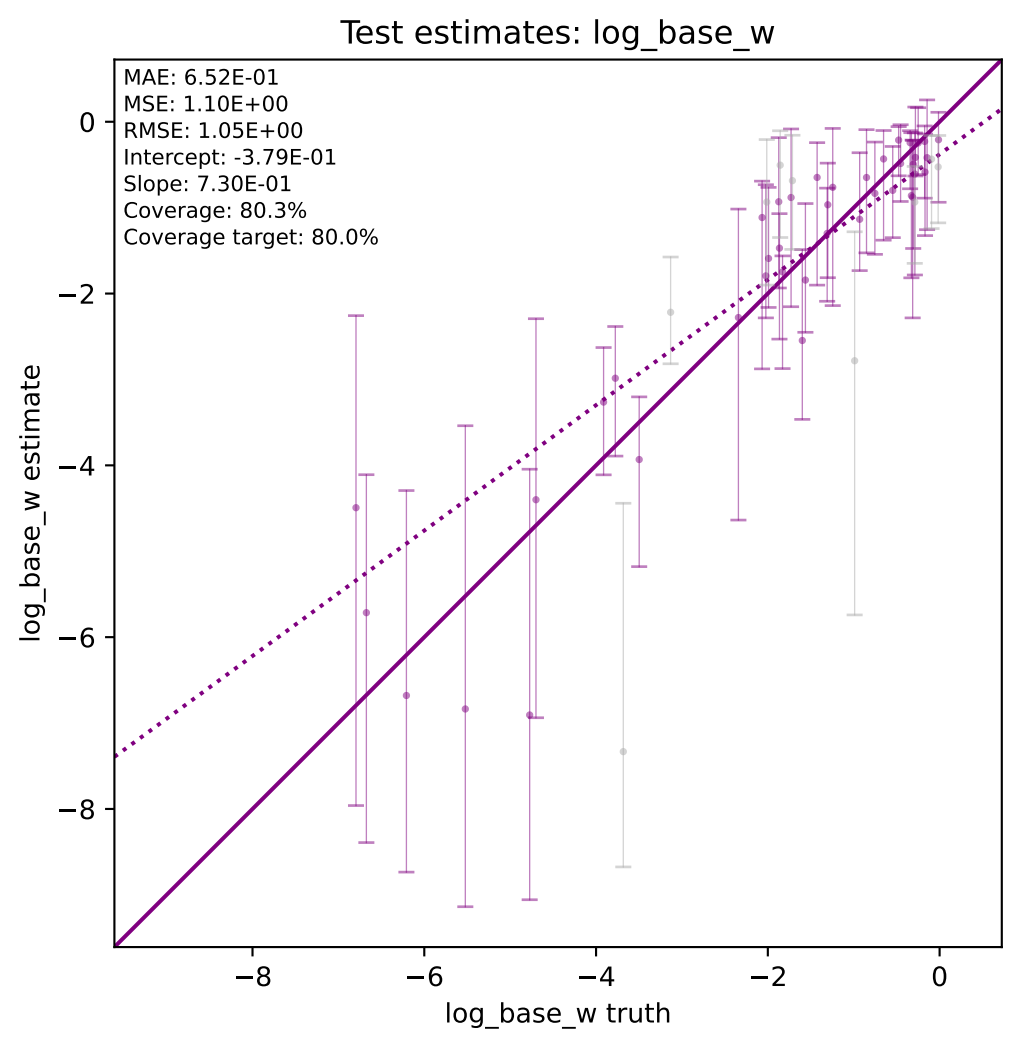

Test estimates: exp_scalar_e
MAE: 1.25E+00
MSE: 2.85E+00
RMSE: 1.69E+00
Intercept: 2.11E+00
Slope: 8.33E-02
Coverage: 80.0%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

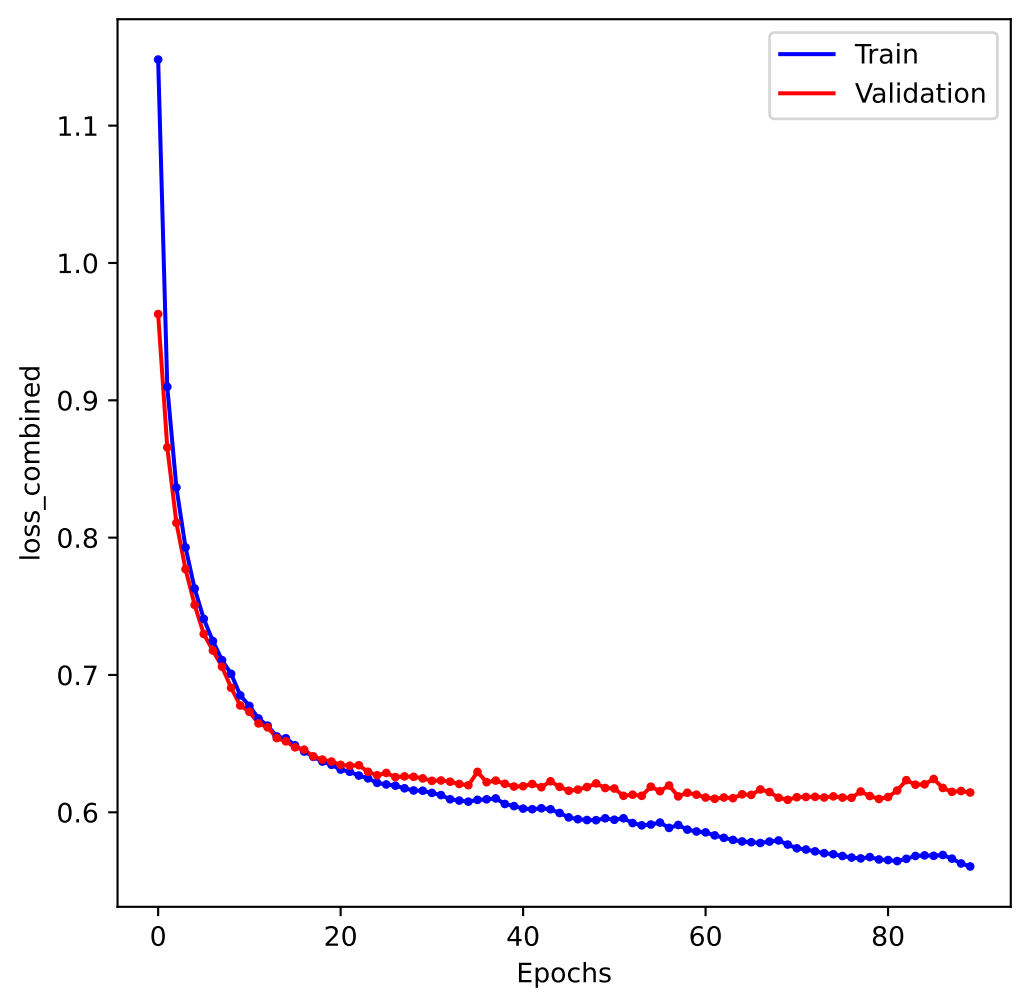

Test estimates: exp_scalar_w
MAE: 1.44E-01
MSE: 3.96E-02
RMSE: 1.99E-01
Intercept: 1.97E-01
Slope: 4.63E-01
Coverage: 79.3%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8

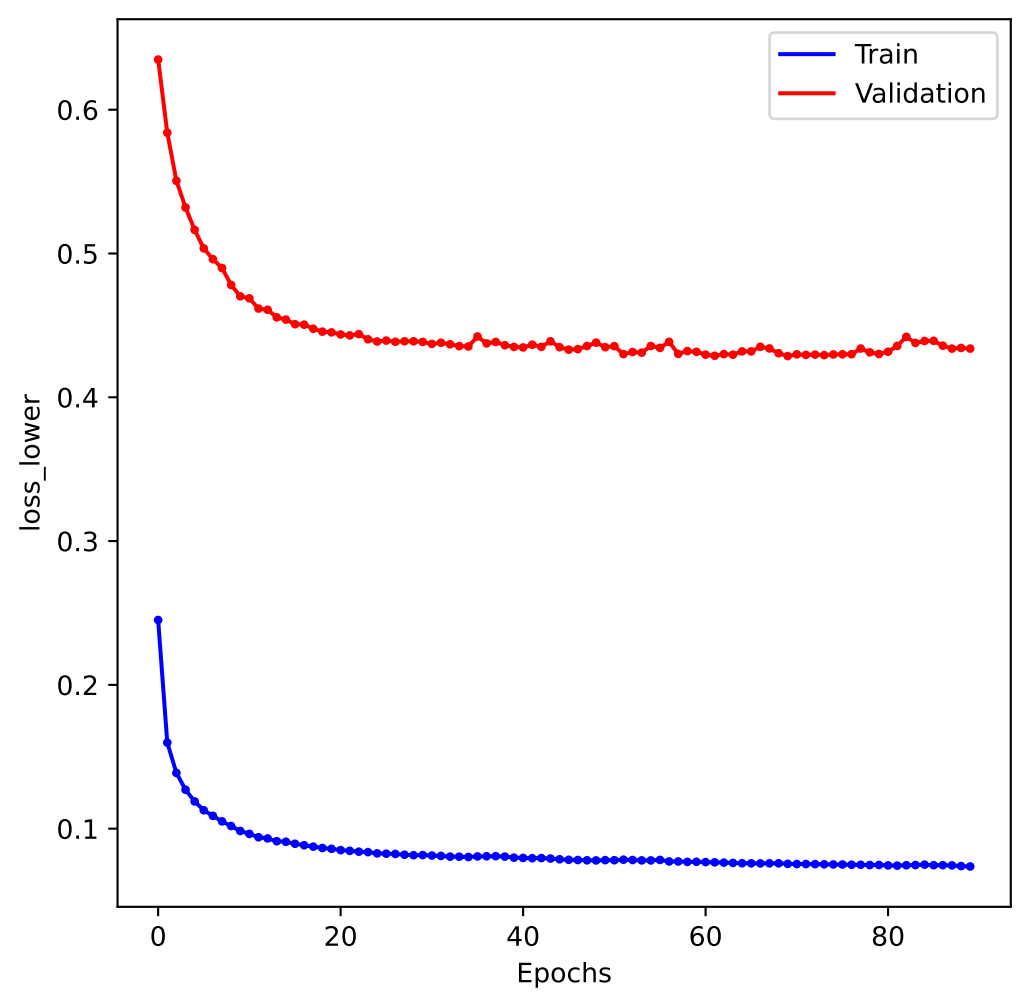

Test estimates: log_base_b
MAE: 7.85E-01
MSE: 1.70E+00
RMSE: 1.30E+00
Intercept: -4.77E-01
Slope: 7.18E-01
Coverage: 80.8%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

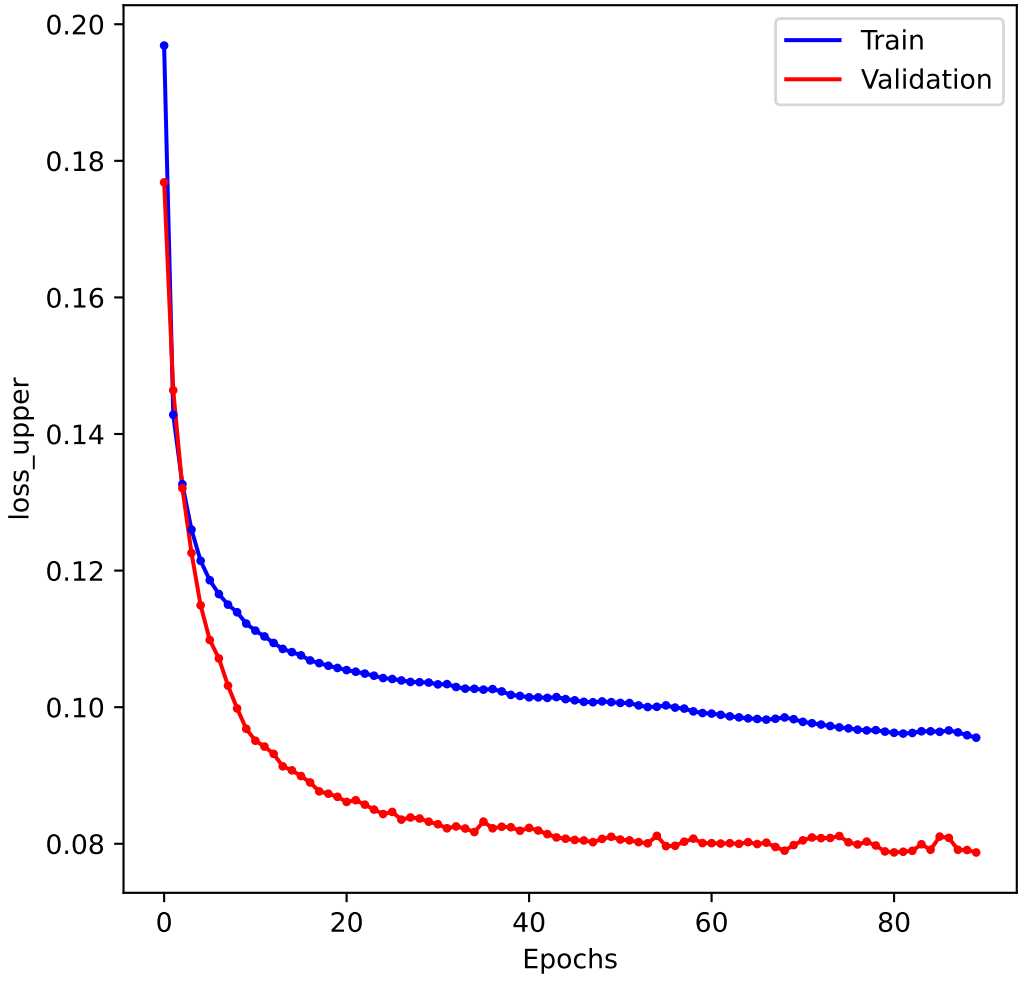

Test estimates: log_base_d
MAE: 4.91E-01
MSE: 4.17E-01
RMSE: 6.46E-01
Intercept: -1.60E-01
Slope: 9.44E-01
Coverage: 81.2%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8

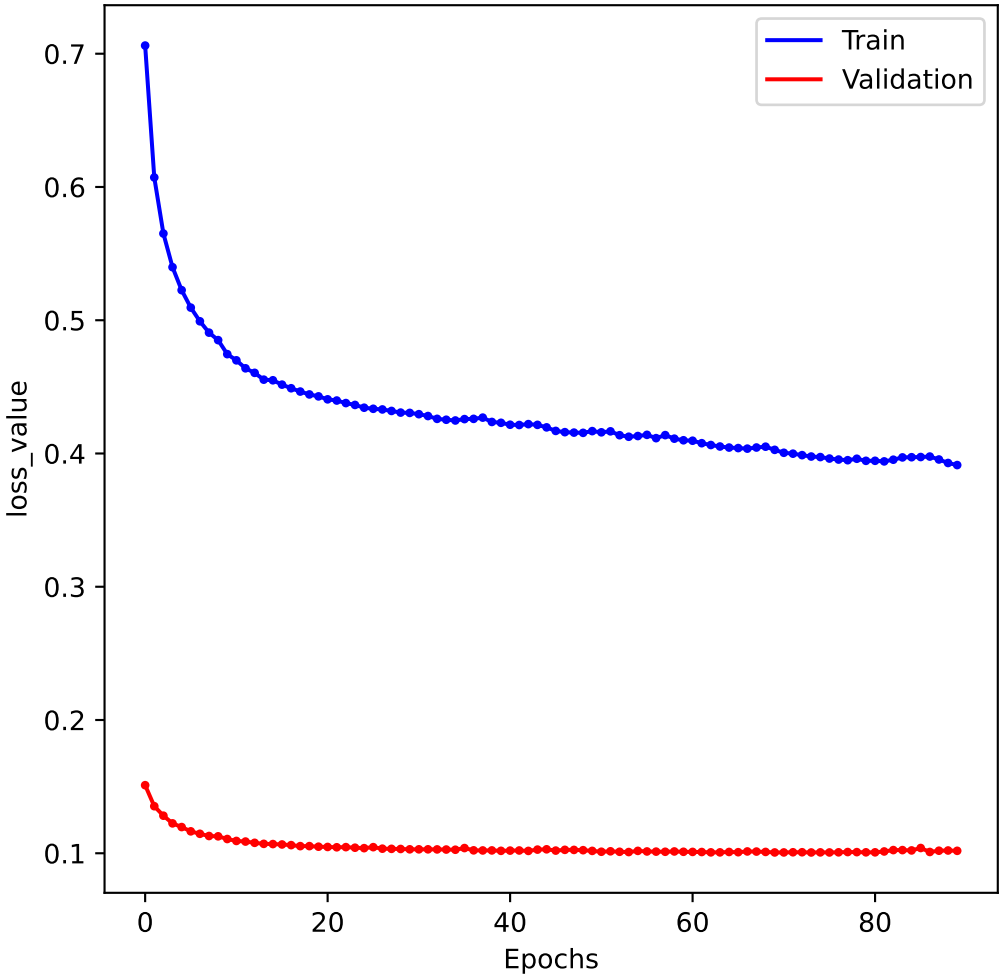

Test estimates: log_base_e
MAE: 7.38E-01
MSE: 8.67E-01
RMSE: 9.31E-01
Intercept: -4.80E-01
Slope: 9.07E-01
Coverage: 79.1%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

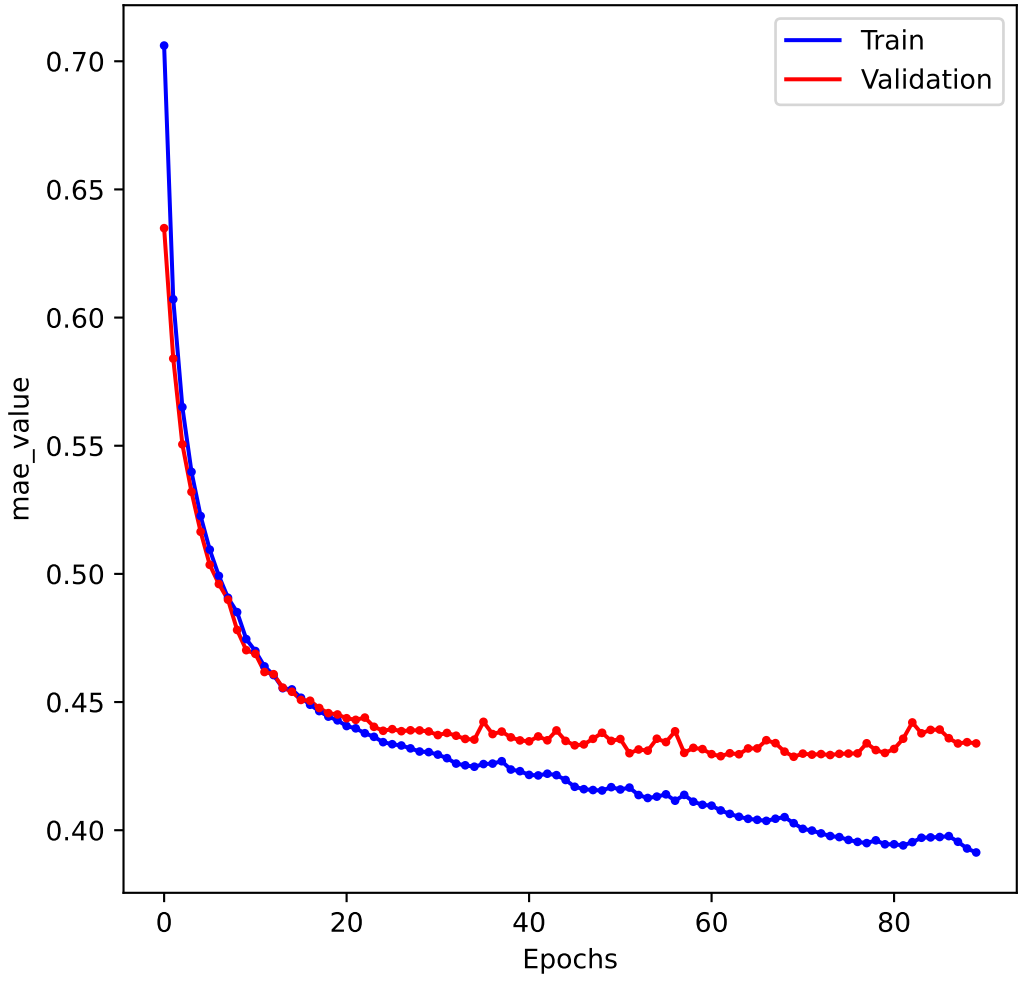

Test estimates: log_base_w
MAE: 6.52E-01
MSE: 1.10E+00
RMSE: 1.05E+00
Intercept: -3.79E-01
Slope: 7.30E-01
Coverage: 80.3%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8

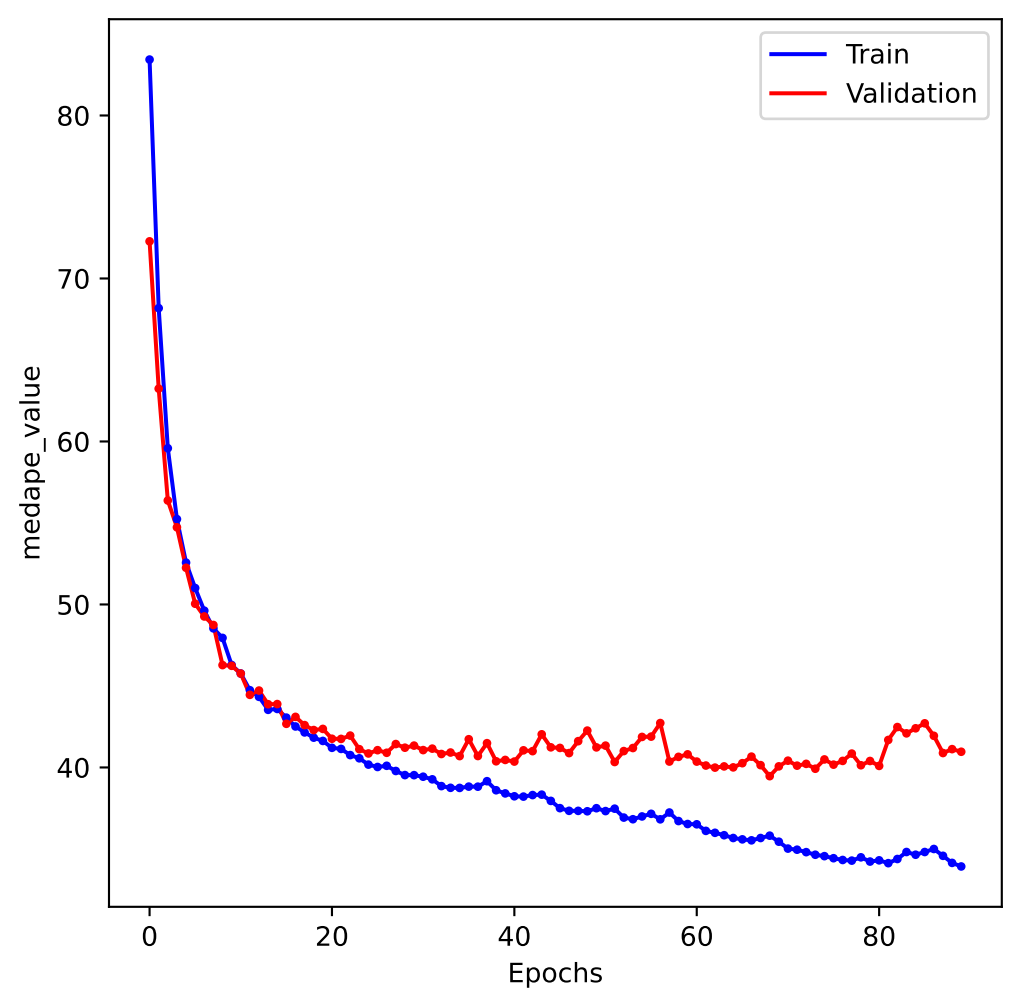

Train
Validation
loss_combined
1.1
1.0
0.9
0.8
0.7
0.6
0
20
40
60
80
Epochs

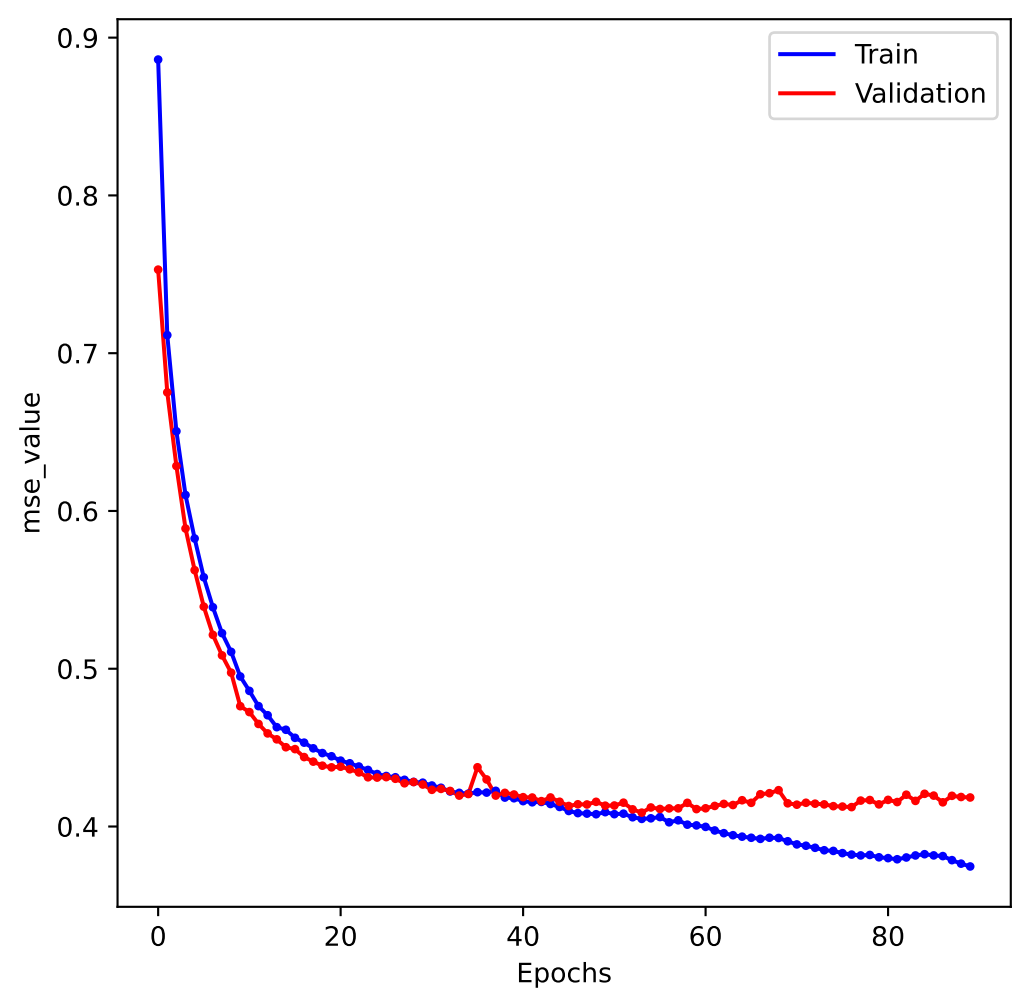

Train
Validation
loss_lower
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
Epochs

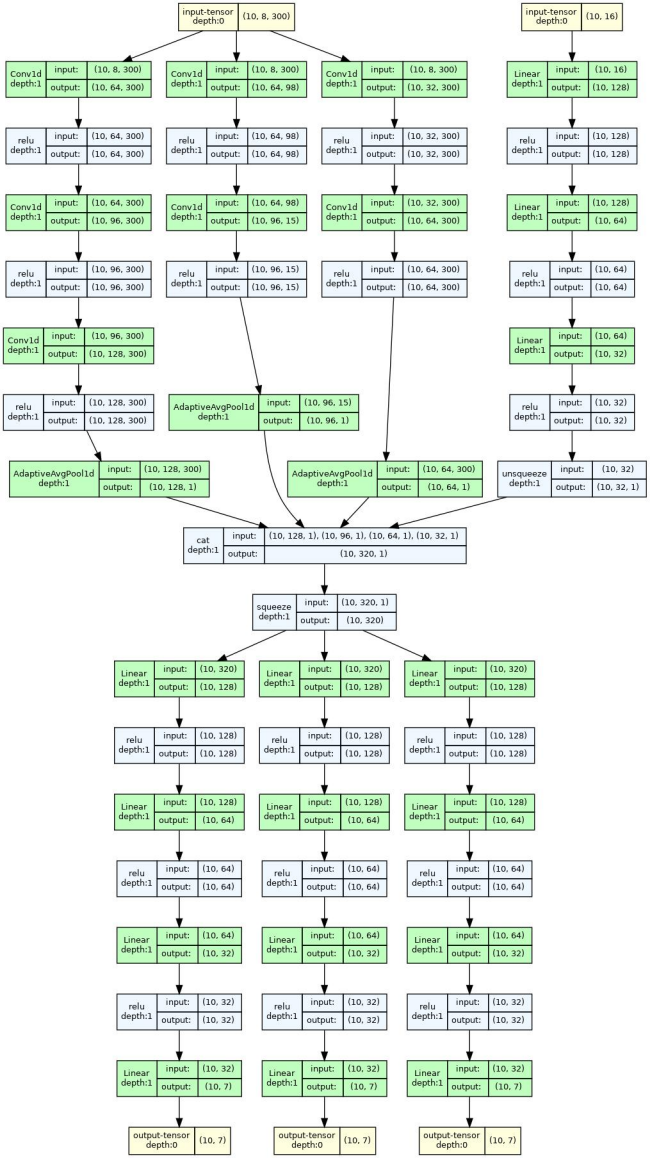

Train
Validation
loss_upper
0.20
0.18
0.16
0.14
0.12
0.10
0.08
0
20
40
60
80
Epochs


Train
Validation
loss_value
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
Epochs


Train
Validation
0.70
0.65
0.60
0.55
0.50
0.45
0.40
mae_value
0
20
40
60
80
Epochs


Train
Validation
medape_value
80
70
60
50
40
0
20
40
60
80
Epochs


Train
Validation
mse_value
Epochs
0.9
0.8
0.7
0.6
0.5
0.4
0
20
40
60
80


input-tensor depth:0 (10, 8, 300)
input-tensor depth:0 (10, 16)
Conv1d depth:1 input: (10, 8, 300) output: (10, 64, 300)
Conv1d depth:1 input: (10, 8, 300) output: (10, 64, 98)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 300)
Linear depth:1 input: (10, 16) output: (10, 128)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 64, 98) output: (10, 64, 98)
relu depth:1 input: (10, 32, 300) output: (10, 32, 300)
relu depth:1 input: (10, 128) output: (10, 128)
Conv1d depth:1 input: (10, 64, 300) output: (10, 96, 300)
Conv1d depth:1 input: (10, 64, 98) output: (10, 96, 15)
Conv1d depth:1 input: (10, 32, 300) output: (10, 64, 300)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 96, 300) output: (10, 96, 300)
relu depth:1 input: (10, 96, 15) output: (10, 96, 15)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 64) output: (10, 64)
Conv1d depth:1 input: (10, 96, 300) output: (10, 128, 300)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 128, 300) output: (10, 128, 300)
AdaptiveAvgPool1d depth:1 input: (10, 96, 15) output: (10, 96, 1)
relu depth:1 input: (10, 32) output: (10, 32)
AdaptiveAvgPool1d depth:1 input: (10, 128, 300) output: (10, 128, 1)
AdaptiveAvgPool1d depth:1 input: (10, 64, 300) output: (10, 64, 1)
unsqueeze depth:1 input: (10, 32) output: (10, 32, 1)
cat depth:1 input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1) output: (10, 320, 1)
squeeze depth:1 input: (10, 320, 1) output: (10, 320)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
Linear depth:1 input: (10, 32) output: (10, 7)
Linear depth:1 input: (10, 32) output: (10, 7)
Linear depth:1 input: (10, 32) output: (10, 7)
output-tensor depth:0 (10, 7)
output-tensor depth:0 (10, 7)
output-tensor depth:0 (10, 7)